

C2 Equation of a circle

Name: _____

Class: _____

Date: _____

Time: **318 minutes**

Marks: **265 marks**

Comments:

EXAM PAPERS PRACTICE

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Q1.

A circle has equation $(x - 2)^2 + (y + 3)^2 = 13$

Find the gradient of the tangent to this circle at the origin.

Circle your answer.

$$-\frac{3}{2}$$

$$-\frac{2}{3}$$

$$\frac{2}{3}$$

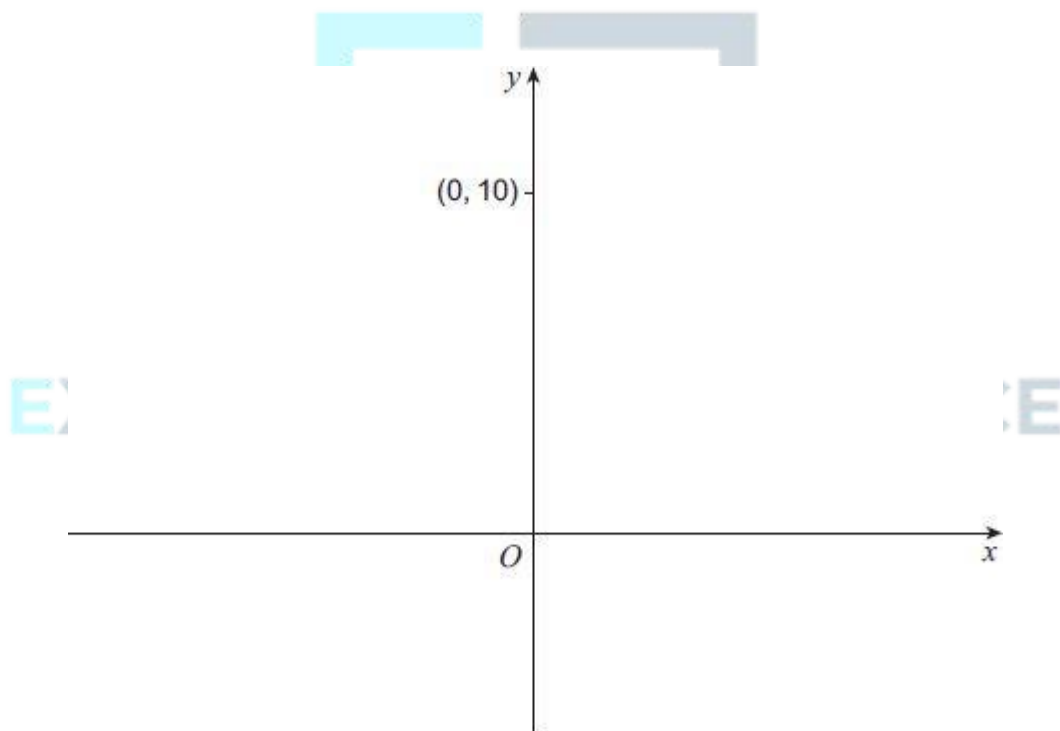
$$\frac{3}{2}$$

(Total 1 mark)

Q2.

A circle of radius 6 passes through the points $(0, 0)$ and $(0, 10)$.

(a) Sketch the two possible positions of the circle.



(1)

(b) Find the equations of the two circles.

(3)

(Total 4 marks)

Q3.

Three points A , B and C have coordinates $A(8, 17)$, $B(15, 10)$ and $C(-2, -7)$

(a) Show that angle ABC is a right angle.

(3)

(b) A , B and C lie on a circle.

(i) Explain why AC is a diameter of the circle.

(1)

(ii) Determine whether the point $D(-8, -2)$ lies inside the circle, on the circle or outside the circle.

Fully justify your answer.

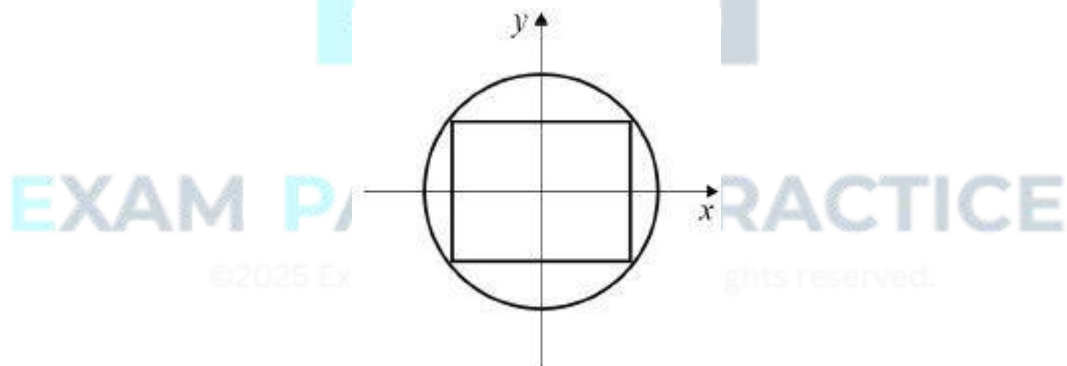
(4)

(Total 8 marks)

Q4.

A company is designing a logo. The logo is a circle of radius 4 inches with an inscribed rectangle. The rectangle must be as large as possible.

The company models the logo on an x - y plane as shown in the diagram.



Use calculus to find the maximum area of the rectangle.

Fully justify your answer.

(Total 10 marks)

Q5.

A circle has equation $(x - 4)^2 + (y + 4)^2 = 9$

What is the area of the circle?

Circle your answer.

3π

9π

16π

81π

(Total 1 mark)

Q6.

The circle with equation $(x - 7)^2 + (y + 2)^2 = 5$ has centre C .

- (a) (i) Write down the radius of the circle.

(1)

- (ii) Write down the coordinates of C .

(1)

- (b) The point $P(5, -1)$ lies on the circle.

Find the equation of the tangent to the circle at P , giving your answer in the form $y = mx + c$

(4)

- (c) The point $Q(3, 3)$ lies outside the circle and the point T lies on the circle such that QT is a tangent to the circle. Find the length of QT .

(4)

(Total 10 marks)

Q7.

A circle with centre C has equation $x^2 + y^2 + 8x - 12y = 12$

- (a) Find the coordinates of C and the radius of the circle.

(3)

- (b) The points P and Q lie on the circle.
The origin is the midpoint of the chord PQ .
Show that PQ has length $n\sqrt{3}$, where n is an integer.

(5)

(Total 8 marks)

Q8.

A circle with centre $C(-3, 2)$ has equation

$$x^2 + y^2 + 6x - 4y = 12$$

- (a) Find the y -coordinates of the points where the circle crosses the y -axis.

(3)

- (b) Find the radius of the circle.

(3)

- (c) The point $P(2, 5)$ lies outside the circle.

- (i) Find the length of CP , giving your answer in the form $\sqrt{n}$, where n is an integer.

(2)

- (ii) The point Q lies on the circle so that PQ is a tangent to the circle. Find the length of PQ .

(2)

(Total 10 marks)

Q9.

A circle C has equation

$$x^2 + y^2 - 10x + 14y + 25 = 0$$

- (a) Write the equation of C in the form

$$(x - a)^2 + (y - b)^2 = k$$

where a , b and k are integers.

(3)

- (b) Hence, for the circle C , write down:

- (i) the coordinates of its centre;

(1)

- (ii) its radius.

(1)

- (c) (i) Sketch the circle C .

(2)

- (ii) Write down the coordinates of the point on C that is furthest away from the x -axis.

(2)

- (d) Given that k has the same value as in part (a), describe geometrically the transformation which maps the circle with equation $(x + 1)^2 + y^2 = k$ onto the circle C .

(3)

(Total 12 marks)

Q10.

A circle with centre C has equation $x^2 + y^2 + 14x - 10y + 49 = 0$.

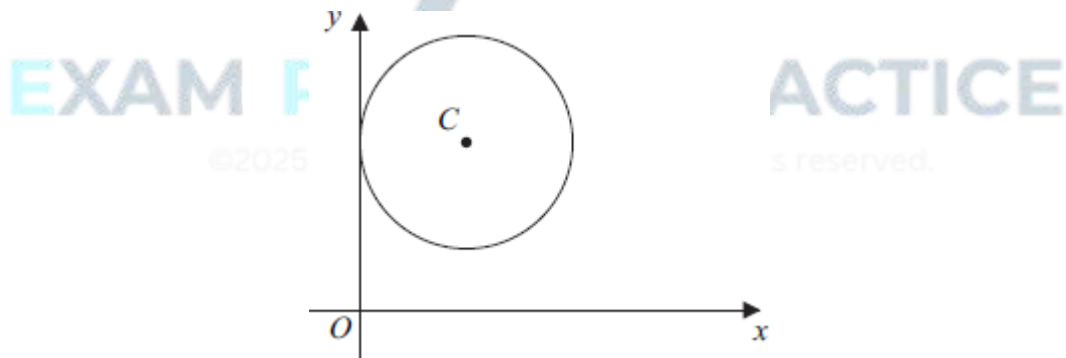
- (a) Express this equation in the form

$$(x - a)^2 + (y - b)^2 = r^2$$

- (b) Write down:
- (i) the coordinates of C ; (3)
 - (ii) the radius of the circle. (2)
- (c) Sketch the circle. (2)
- (d) A line has equation $y = kx + 6$, where k is a constant.
- (i) Show that the x -coordinates of any points of intersection of the line and the circle satisfy the equation $(k^2 + 1)x^2 + 2(k + 7)x + 25 = 0$. (2)
 - (ii) The equation $(k^2 + 1)x^2 + 2(k + 7)x + 25 = 0$ has equal roots. Show that $12k^2 - 7k - 12 = 0$ (3)
 - (iii) Hence find the values of k for which the line is a tangent to the circle. (2)
- (Total 14 marks)

Q11.

The circle with centre $C(5, 8)$ touches the y -axis, as shown in the diagram.



- (a) Express the equation of the circle in the form
- $$(x - a)^2 + (y - b)^2 = k$$
- (2)
- (b) (i) Verify that the point $A(2, 12)$ lies on the circle. (1)
- (ii) Find an equation of the tangent to the circle at the point A , giving your answer in the form $sx + ty + u = 0$, where s , t and u are integers.

(5)

(c) The points P and Q lie on the circle, and the mid-point of PQ is $M(7, 12)$.

(i) Show that the length of CM is $n\sqrt{5}$, where n is an integer.

(2)

(ii) Hence find the area of triangle PCQ .

(3)

(Total 13 marks)

Q12.

A circle has centre $C(-3, 1)$ and radius $\sqrt{13}$.

(a) (i) Express the equation of the circle in the form

$$(x - a)^2 + (y - b)^2 = k$$

(2)

(ii) Hence find the equation of the circle in the form

$$x^2 + y^2 + mx + ny + p = 0$$

where m , n and p are integers.

(3)

(b) The circle cuts the y -axis at the points A and B . Find the distance AB .

(3)

(c) (i) Verify that the point $D(-5, -2)$ lies on the circle.

(1)

(ii) Find the gradient of CD .

(2)

(iii) Hence find an equation of the tangent to the circle at the point D .

(2)

(Total 13 marks)

Q13.

A circle has centre $C(3, -8)$ and radius 10.

(a) Express the equation of the circle in the form

$$(x - a)^2 + (y - b)^2 = k$$

(2)

(b) Find the x -coordinates of the points where the circle crosses the x -axis.

(3)

- (c) The tangent to the circle at the point A has gradient $\frac{5}{2}$. Find an equation of the line CA , giving your answer in the form $rx + sy + t = 0$, where r , s and t are integers. (3)

- (d) The line with equation $y = 2x + 1$ intersects the circle.

- (i) Show that the x -coordinates of the points of intersection satisfy the equation

$$x^2 + 6x - 2 = 0$$

(3)

- (ii) Hence show that the x -coordinates of the points of intersection are of the form $m \pm \sqrt{n}$, where m and n are integers.

(2)

(Total 13 marks)

Q14.

A circle with centre C has equation $x^2 + y^2 - 4x + 12y + 15 = 0$.

- (a) Find:

- (i) the coordinates of C ;

(2)

- (ii) the radius of the circle.

(2)

- (b) Explain why the circle lies entirely below the x -axis.

(2)

- (c) The point P with coordinates $(5, k)$ lies outside the circle.

- (i) Show that $PC^2 = k^2 + 12k + 45$.

(2)

- (ii) Hence show that $k^2 + 12k + 20 > 0$.

(1)

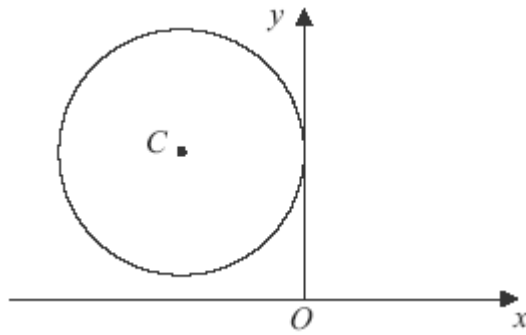
- (iii) Find the possible values of k .

(4)

(Total 13 marks)

Q15.

A circle with centre $C(-5, 6)$ touches the y -axis, as shown in the diagram.



- (a) Find the equation of the circle in the form

$$(x - a)^2 + (y - b)^2 = r^2$$

(3)

- (b) (i) Verify that the point $P(-2, 2)$ lies on the circle.

(1)

- (ii) Find an equation of the normal to the circle at the point P .

(3)

- (iii) The mid-point of PC is M . Determine whether the point P is closer to the point M or to the origin O .

(4)

(Total 11 marks)

Q16.

A circle with centre C has equation $x^2 + y^2 - 6x + 10y + 9 = 0$.

- (a) Express this equation in the form

$$(x - a)^2 + (y - b)^2 = r^2$$

(3)

- (b) Write down:

- (i) the coordinates of C ;
 (ii) the radius of the circle.

(2)

- (c) The point D has coordinates $(7, -2)$.

- (i) Verify that the point D lies on the circle.

(1)

- (ii) Find an equation of the normal to the circle at the point D , giving your answer in the form $mx + ny = p$, where m , n and p are integers.

(3)

- (d) (i) A line has equation $y = kx$. Show that the x -coordinates of any points of intersection of the line and the circle satisfy the equation

$$(k^2 + 1)x^2 + 2(5k - 3)x + 9 = 0$$

(2)

- (ii) Find the values of k for which the equation

$$(k^2 + 1)x^2 + 2(5k - 3)x + 9 = 0$$

has equal roots.

(5)

- (iii) Describe the geometrical relationship between the line and the circle when k takes either of the values found in part (d)(ii).

(1)

(Total 17 marks)

Q17.

A circle with centre C has equation

$$(x - 5)^2 + (y + 12)^2 = 169$$

- (a) Write down:

- (i) the coordinates of C ;

(1)

- (ii) the radius of the circle.

(1)

- (b) (i) Verify that the circle passes through the origin O .

(1)

- (ii) Given that the circle also passes through the points $(10, 0)$ and $(0, p)$, sketch the circle and find the value of p .

(3)

- (c) The point $A(-7, -7)$ lies on the circle.

- (i) Find the gradient of AC .

(2)

- (ii) Hence find an equation of the tangent to the circle at the point A , giving your answer in the form $ax + by + c = 0$, where a , b and c are integers.

(3)

(Total 11 marks)

Q18.

A circle with centre C has equation $x^2 + y^2 - 10y + 20 = 0$.

- (a) By completing the square, express this equation in the form

$$x^2 + (y - b)^2 = k$$

(2)

- (b) Write down:

- (i) the coordinates of C ;

(1)

- (ii) the radius of the circle, leaving your answer in surd form.

(1)

- (c) A line has equation $y = 2x$.

- (i) Show that the x -coordinate of any point of intersection of the line and the circle satisfies the equation $x^2 - 4x + 4 = 0$.

(2)

- (ii) Hence show that the line is a tangent to the circle and find the coordinates of the point of contact, P .

(3)

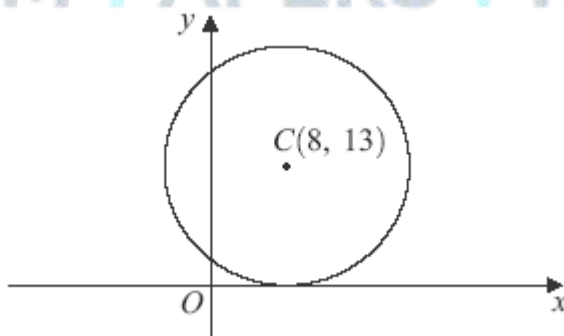
- (d) Prove that the point $Q(-1, 4)$ lies inside the circle.

(2)

(Total 11 marks)

Q19.

The circle S has centre $C(8, 13)$ and touches the x -axis, as shown in the diagram.



- (a) Write down an equation for S , giving your answer in the form

$$(x - a)^2 + (y - b)^2 = r^2$$

(2)

- (b) The point P with coordinates $(3, 1)$ lies on the circle.

- (i) Find the gradient of the straight line passing through P and C .

(1)

- (ii) Hence find an equation of the tangent to the circle S at the point P , giving your answer in the form $ax + by = c$, where a , b and c are integers.

(4)

- (iii) The point Q also lies on the circle S , and the length of PQ is 10. Calculate the shortest distance from C to the chord PQ .

(3)

(Total 10 marks)

Q20.

A circle with centre C has equation $x^2 + y^2 + 2x - 12y + 12 = 0$.

- (a) By completing the square, express this equation in the form

$$(x - a)^2 + (y - b)^2 = r^2$$

(3)

- (b) Write down:

- (i) the coordinates of C ;

(1)

- (ii) the radius of the circle.

(1)

- (c) Show that the circle does **not** intersect the x -axis.

(2)

- (d) The line with equation $x + y = 4$ intersects the circle at the points P and Q .

- (i) Show that the x -coordinates of P and Q satisfy the equation

$$x^2 + 3x - 10 = 0$$

(3)

- (ii) Given that P has coordinates $(2, 2)$, find the coordinates of Q .

(2)

- (iii) Hence find the coordinates of the midpoint of PQ .

(2)

(Total 14 marks)

Q21.

A circle with centre C has equation $(x + 3)^2 + (y - 2)^2 = 25$.

- (a) Write down:

- (i) the coordinates of C ; (2)
 - (ii) the radius of the circle. (1)
 - (b) (i) Verify that the point $N(0, -2)$ lies on the circle. (1)
 - (ii) Sketch the circle. (2)
 - (iii) Find an equation of the normal to the circle at the point N . (3)
 - (c) The point P has coordinates $(2, 6)$.
 - (i) Find the distance PC , leaving your answer in surd form. (2)
 - (ii) Find the length of a tangent drawn from P to the circle. (3)
- (Total 14 marks)**

Q22.

A circle with centre C has equation $x^2 + y^2 - 8x + 6y = 11$.

- (a) By completing the square, express this equation in the form

$$(x - a)^2 + (y - b)^2 = r^2$$

- (b) Write down:

- (i) the coordinates of C ; (1)
- (ii) the radius of the circle. (1)
- (c) The point O has coordinates $(0, 0)$.
- (i) Find the length of CO . (2)
- (ii) Hence determine whether the point O lies inside or outside the circle, giving a reason for your answer. (2)

(Total 9 marks)

Q23.

A circle has equation $x^2 + y^2 - 4x - 14 = 0$.

(a) Find:

(i) the coordinates of the centre of the circle;

(3)

(ii) the radius of the circle in the form $p\sqrt{2}$, where p is an integer.

(3)

(b) A chord of the circle has length 8. Find the perpendicular distance from the centre of the circle to this chord.

(3)

(c) A line has equation $y = 2k - x$, where k is a constant.

(i) Show that the x -coordinate of any point of intersection of the line and the circle satisfies the equation

$$x^2 - 2(k + 1)x + 2k^2 - 7 = 0$$

(3)

(ii) Find the values of k for which the equation

$$x^2 - 2(k + 1)x + 2k^2 - 7 = 0$$

has equal roots.

(4)

(iii) Describe the geometrical relationship between the line and the circle when k takes either of the values found in part (c)(ii).

(1)

(Total 17 marks)

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Q24.

A circle has equation $x^2 + y^2 - 12x - 6y + 20 = 0$.

(a) By completing the square, express the equation in the form

$$(x - a)^2 + (y - b)^2 = r^2$$

(3)

(b) Write down:

(i) the coordinates of the centre of the circle;

(1)

(ii) the radius of the circle.

(1)

(c) The line with equation $y = x + 4$ intersects the circle at the points P and Q .

(i) Show that the x -coordinates of P and Q satisfy the equation

$$x^2 - 5x + 6 = 0$$

(2)

(ii) Find the coordinates of P and Q .

(4)

(Total 11 marks)

Q25.

A circle has centre C (2, -1) and radius 5. The point P has coordinates (6, 2).

(a) Write down an equation of the circle.

(3)

(b) Verify that the point P lies on the circle.

(2)

(c) Find the gradient of the line CP .

(2)

(d) (i) Find the gradient of a line which is perpendicular to CP .

(2)

(ii) Hence find an equation for the tangent to the circle at the point P .

(1)

(Total 10 marks)